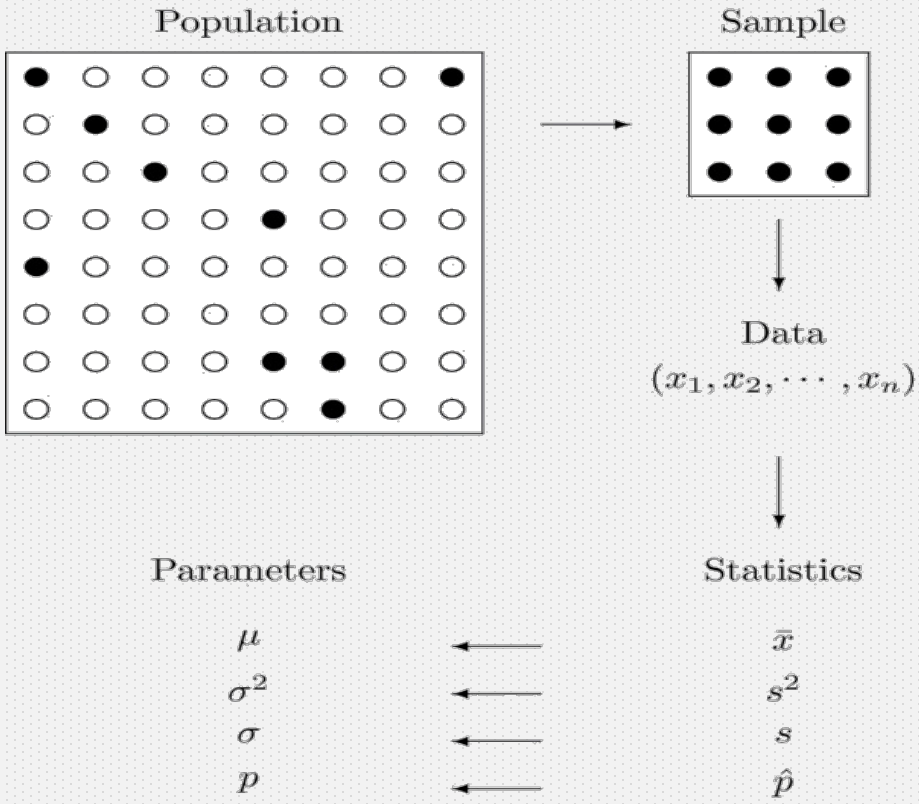
DECEMBER	TOPIC
4	ESTIMATION

INTRODUCTION

The purpose of statistics is to estimate an unknown parameter of a population and to assess the reliability of this estimation. Estimation falls into two categories: **point estimation** and **interval estimation.**



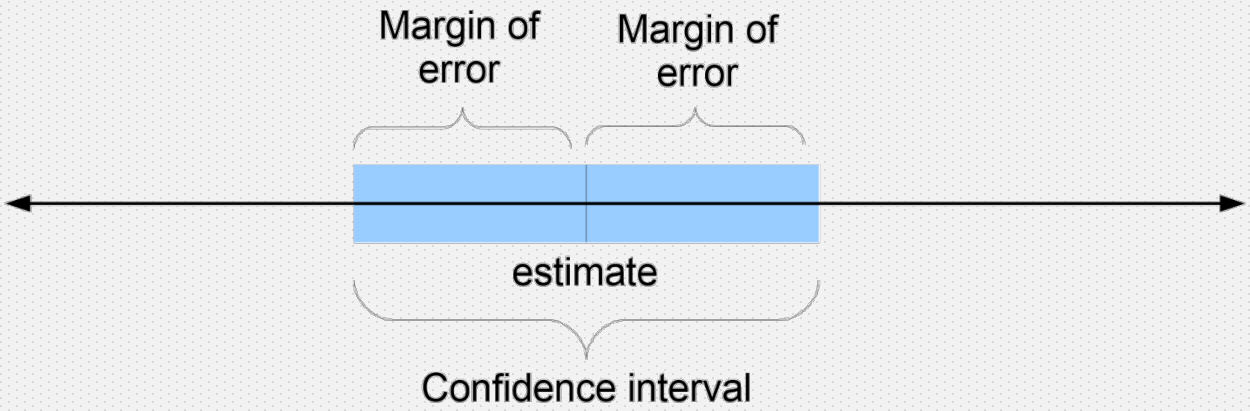
INTRODUCTION

Point Estimation: using the data to calculate a single estimate of the parameter of interest. For example, we often use te sample mean $\bar{x}$ to estimate the population mean μ.

Interval Estimation: provides a range of values (an interval) that may contain the unknown parameter (such as the population mean).

CONFIDENCE INTERVAL

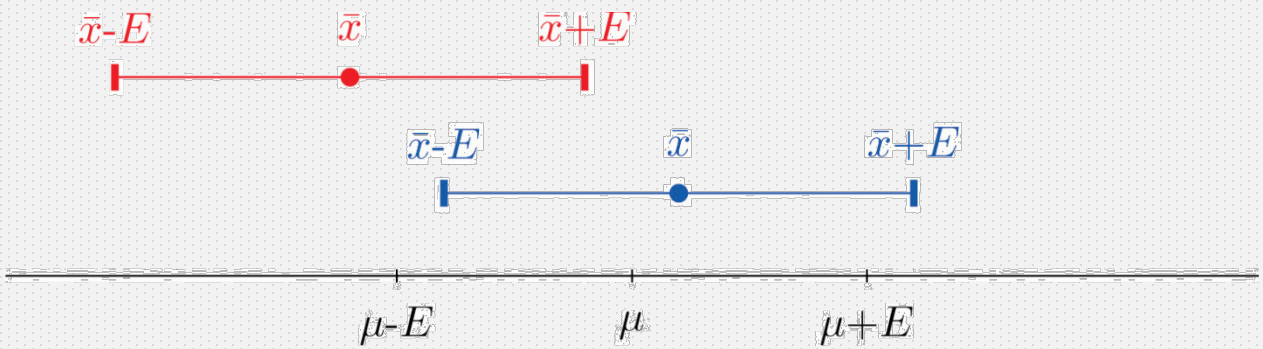
The confidence level is the probability that intervals determined by these methods will contain the parameter. A confidence interval is the range of values determined from a sample statistic and the specified confidence level.



The common confidence intervals use 90%, 95%, or 99% confidence levels.

DEVELOPMENT OF CONFIDENCE INTERVAL

Using the sample mean $\bar{x}$, to estimate the population mean μ.



$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}}$$

DEVELOPMENT OF CONFIDENCE INTERVAL

Whenever the population standard deviation σ is known and either the population is normally distributed or $n \geq 30$, the Central Limit Theorem guarantees the sample mean is normally distributed or:

$$-z < \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} < z$$

DEVELOPMENT OF CONFIDENCE INTERVAL

$$-z < \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} < z$$

$$-z\sigma_{\bar{x}} < \bar{x} - \mu < z\sigma_{\bar{x}}$$

$$-\bar{x} - z\sigma_{\bar{x}} < -\mu < z\sigma_{\bar{x}} - \bar{x}$$

$$-(-\bar{x} - z\sigma_{\bar{x}}) > -(-\mu) > -(z\sigma_{\bar{x}} - \bar{x})$$

$$\bar{x} + z\sigma_{\bar{x}} > \mu > \bar{x} - z\sigma_{\bar{x}}$$

$$\bar{x} - z\sigma_{\bar{x}} < \mu < \bar{x} + z\sigma_{\bar{x}}$$

Recall: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

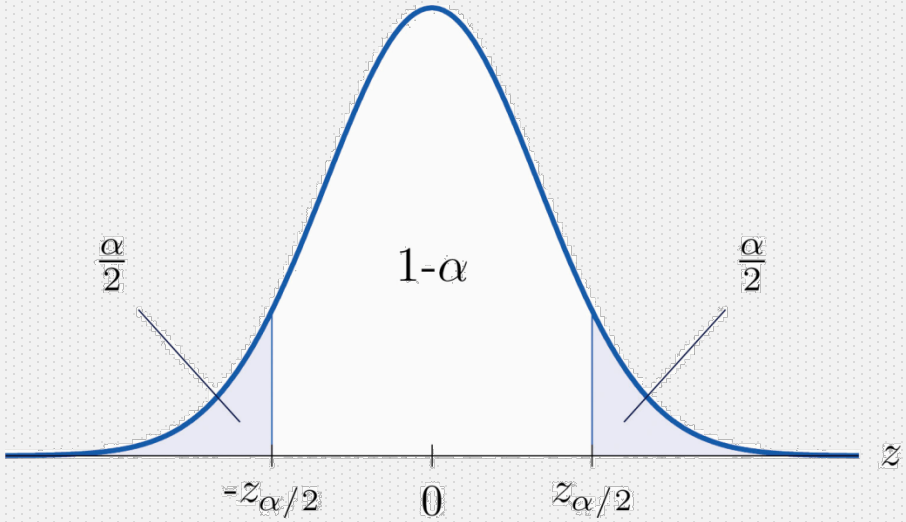
$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}}$$

DEVELOPMENT OF CONFIDENCE INTERVAL

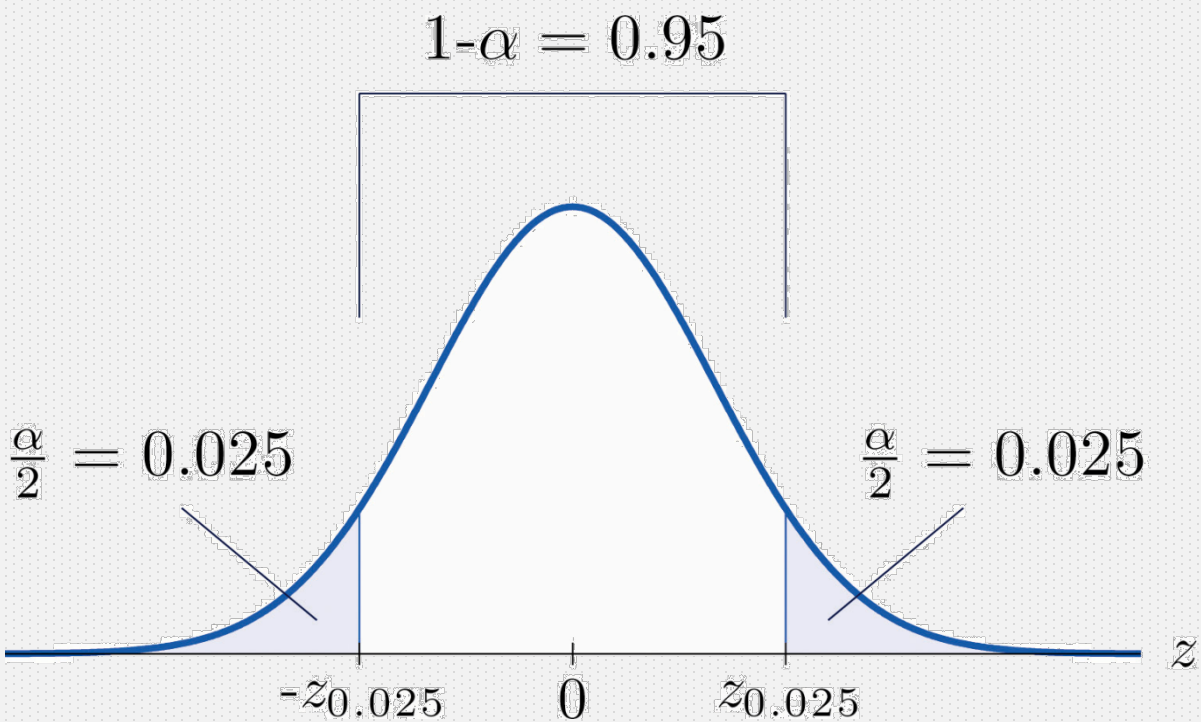
$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}}$$

$\bar{x}$: is the point estimate and the center of the confidence interval.

z: is the confidence coefficient, the number of multiples of the standard error needed to construct an interval estimate of the correct width to have a level of confidence $1 - \alpha$



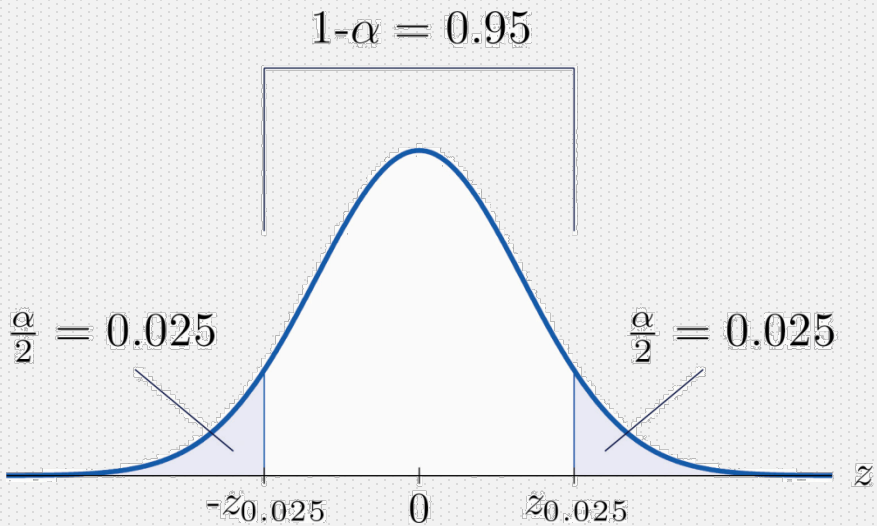
95 % CONFIDENCE INTERVAL



In order to find the value of $z_{0.025}$ we have to find the entry of $0.025 + 0.95 = 0.975$ from the Z table. Same applies to $-z_{0.025}$

Recall: http://yunus.hacettepe.edu.tr/~gertunc/upload/MAD213_19OCT.pdf

95 % CONFIDENCE INTERVAL

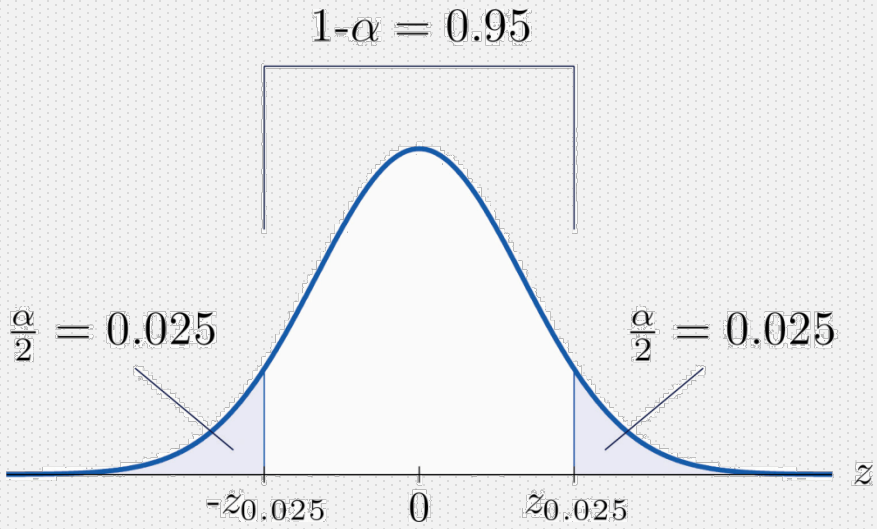


$$z_{0.025} = 1.96$$

PROBABILITY CONTENT FROM z to ∞

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

95 % CONFIDENCE INTERVAL



$$z_{0.025} = 1.96$$

For 95% Confidence Interval:

$$\bar{x} - 1.96\,\sigma_{\bar{x}} < \mu < \bar{x} + 1.96\,\sigma_{\bar{x}}$$

EXAMPLE

120 students are registered to MAD213 course. First midterm average score is obtained as 75. Standard deviation of the population is 16.

Find the 95% confidence interval estimate for the mean reading scores of all students of MAD213 course.

$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}}$$

SOLUTION

n: 120

$\bar{x}$: 75

σ: 16

z: 1.96 for 95% CI

Constructing 95% CI:

$$75 - 1.96\frac{16}{\sqrt{120}} < \mu < 75 + 1.96\frac{16}{\sqrt{120}}$$

$$75 - 2.86 < \mu < 75 + 2.86$$

$$\mathbf{72.14 < \mu < 77.86}$$

"I am 95 % confident that, midterm results lies between 72.14 and 77.86"

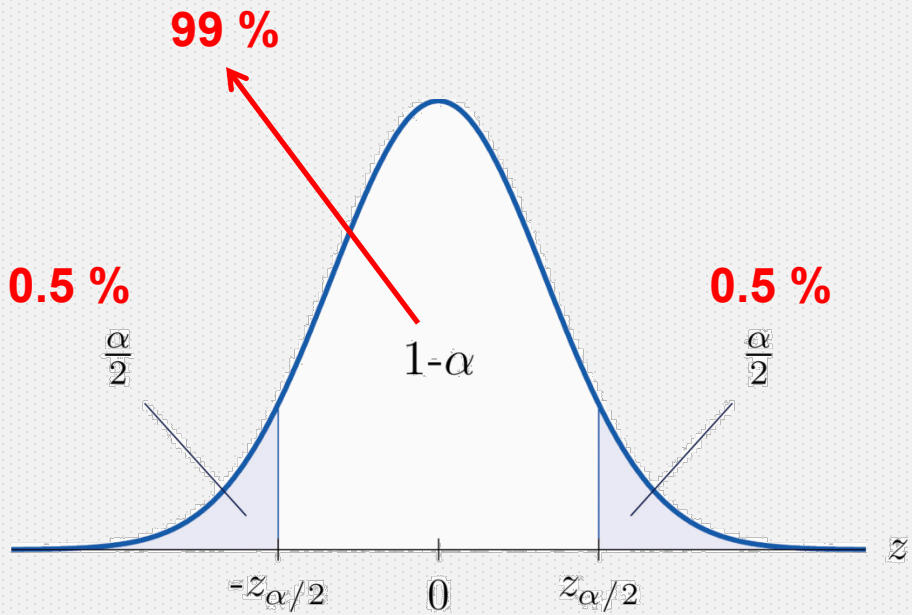
EXAMPLE

120 students are registered to MAD213 course. First midterm average score is obtained as 75. Standard deviation of the population is 16.

Find the 99% confidence interval estimate for the mean reading scores of all students of MAD213 course.

$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}}$$

99 % CONFIDENCE INTERVAL



From table we have to find $z_{0.005}$ that is

$$0.005 + 0.99 = 0.995$$

$$z_{0.005} = 2.575$$

PROBABILITY CONTENT FROM z to ∞

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
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2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

SOLUTION

n: 120

$\bar{x}$: 75

σ: 16

z: 2.575 for 99% CI

Constructing 99% CI:

$$75 - 2.575\frac{16}{\sqrt{120}} < \mu < 75 + 2.575\frac{16}{\sqrt{120}}$$

$$75 - 4.8 < \mu < 75 + 4.8$$

$$\mathbf{70.2 < \mu < 79.8}$$

"I am 99 % confident that, midterm results lies between 70.2 and 79.8"

KEY TAKEAWAYS

- A confidence interval for a population mean is an estimate of the population mean together with an indication of reliability.
- The CI are constructed entirely from the sample data (or sample data and the population standard deviation, when it is known)
- There are different formulas for a CI based on the sample size and whether or not the population standard deviation is known.

SMALL SAMPLE ESTIMATION

The confidence interval formulas in the previous section are based on the Central Limit Theorem, the statement that for large samples $\bar{x}$ is normally distributed with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

When the population mean μ is estimated with a small sample ($n < 30$), the Central Limit Theorem does not apply.

SMALL SAMPLE ESTIMATION

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When the population mean μ is estimated with a small sample ($n < 30$), the Central Limit Theorem does not apply.

CRITICAL VALUES OF $t_{a/2}$

df	0.250	0.200	0.150	0.100	0.050	0.025	0.010	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390

$$\bar{x} - t_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2}\frac{s}{\sqrt{n}}$$

Degree of freedom $\boldsymbol{df} = n - 1$

EXAMPLE

A sample of size 12 from a normally distributed population has sample mean 35 and sample standard deviation 14. Construct 90 % CI for the population mean, and interpret its meaning.

$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}}$$

EXAMPLE

$$n = 12$$
$$degree\ of\ freedom = n - 1 = 12 - 1 = 11$$
$$\bar{x} = 35$$
$$s = 14$$

90%CI means that $\frac{\alpha}{2} = \frac{0.1}{2} = 0.05 \quad t_{\alpha/2} = t_{0.05}$

$$\bar{x} - t_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2}\frac{s}{\sqrt{n}}$$

$$35 - t_{\alpha/2}\frac{14}{\sqrt{12}} < \mu < 35 + t_{\alpha/2}\frac{14}{\sqrt{12}}$$

CRITICAL VALUES OF $t_{a/2}$

df	0.250	0.200	0.150	0.100	0.050	0.025	0.010	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	$\boxed{1.761}$	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390

$$35 - t_{0.05}\frac{14}{\sqrt{12}} < \mu < 35 + t_{0.05}\frac{14}{\sqrt{12}}$$

$$\boldsymbol{df} = 11$$

$$35 - 1.761\frac{14}{\sqrt{12}} < \mu < 35 + 1.761\frac{14}{\sqrt{12}}$$

$$27.88 < \mu < 42.12$$

KEY TAKEAWAYS

- In selecting the correct formula for construction of a confidence interval for a population mean ask two questions: is the population standard deviation σ known or unknown, and is the sample large or small?
- Confidence intervals can be constructed with small samples only if the population is normal.